

НОВЫЕ ТЕХНОЛОГИИ / НАУКА

По словам IBM, квантовые компьютеры становятся все сложнее проверить. Это прогресс.

По данным IBM, три эксперимента демонстрируют квантовое преимущество перед проверенными классическими методами. Но как исследователям доверять результатам, которые не может проверить классический компьютер?

30 июля 2026 года, 6:00 [Фредерик Лардинуа](#)



IBM и ее партнеры утверждают, что квантовые компьютеры уже достигли такого уровня, когда они могут выполнять вычисления, которые невозможно воспроизвести и проверить с помощью классических методов на современных компьютерах.

Хотя это, безусловно, признак прогресса, учитывая, что до появления полезных квантовых компьютеров всегда оставалось пять лет, сейчас мы

В конце концов, результат, который никто не может проверить, не так уж полезен.

Как отмечает группа исследователей из IBM и Чикагского университета в одной из трех статей, опубликованных IBM совместно с группой партнеров в четверг, чтобы поверить в то, что квантовые компьютеры могут «обеспечить экспоненциальное сокращение времени выполнения по сравнению с классическими вычислениями для определенных задач», нам нужно верить в результаты.

«Помимо теоретико-множественной основы, [...] масштабируемая демонстрация квантового превосходства должна соответствовать еще двум критериям, — пишут исследователи. — Во-первых, увеличение размера схемы должно сопровождаться соответствующим подавлением шума, чтобы эксперимент не поддавался классическому моделированию по мере масштабирования [13]. Во-вторых, нужно быть уверенным, что эти сложные вычисления были выполнены с высокой точностью даже в условиях шума».

Компания IBM и ее партнеры объявили, что нашли новые способы повысить достоверность этих результатов. Исследователи описывают расчеты, которые превосходят ведущие классические методы, которые они тестировали, но при этом позволяют оценить результаты.

Хотя исследованиями руководили IBM и Чикагский университет, в различных аспектах их поддерживали компании Qedma, RIKEN, BlueQubit и Algorithmiq.

ТРЕНДОВЫЕ ИСТОРИИ

1. [IBM says quantum computers are getting harder to verify. That's progress.](#)
2. [Microsoft Makes Quantum Computing Breakthrough With New Chip](#)
3. [Why You Can't Debug a Running Quantum Computer Program](#)
4. [Ladybird: That Rare Breed of Browser Based on Web Standards](#)
5. [A Developer's Guide to ActivityPub and the Fediverse](#)

Только в одной из статей эксперимент прямо назван протоколом с квантовым преимуществом. В пресс-релизе IBM утверждается, что работа, описанная в этой статье, является «демонстрацией квантовых вычислений, которые соответствуют фундаментальным критериям квантового преимущества: выполнение вычислений, недоступных для ведущих классических методов моделирования, при сохранении уверенности в том, что вычисления дали точные результаты».

В двух других статьях более осторожно говорится о выходе за рамки проверенных классических методов и установлении достоверности квантовых результатов.

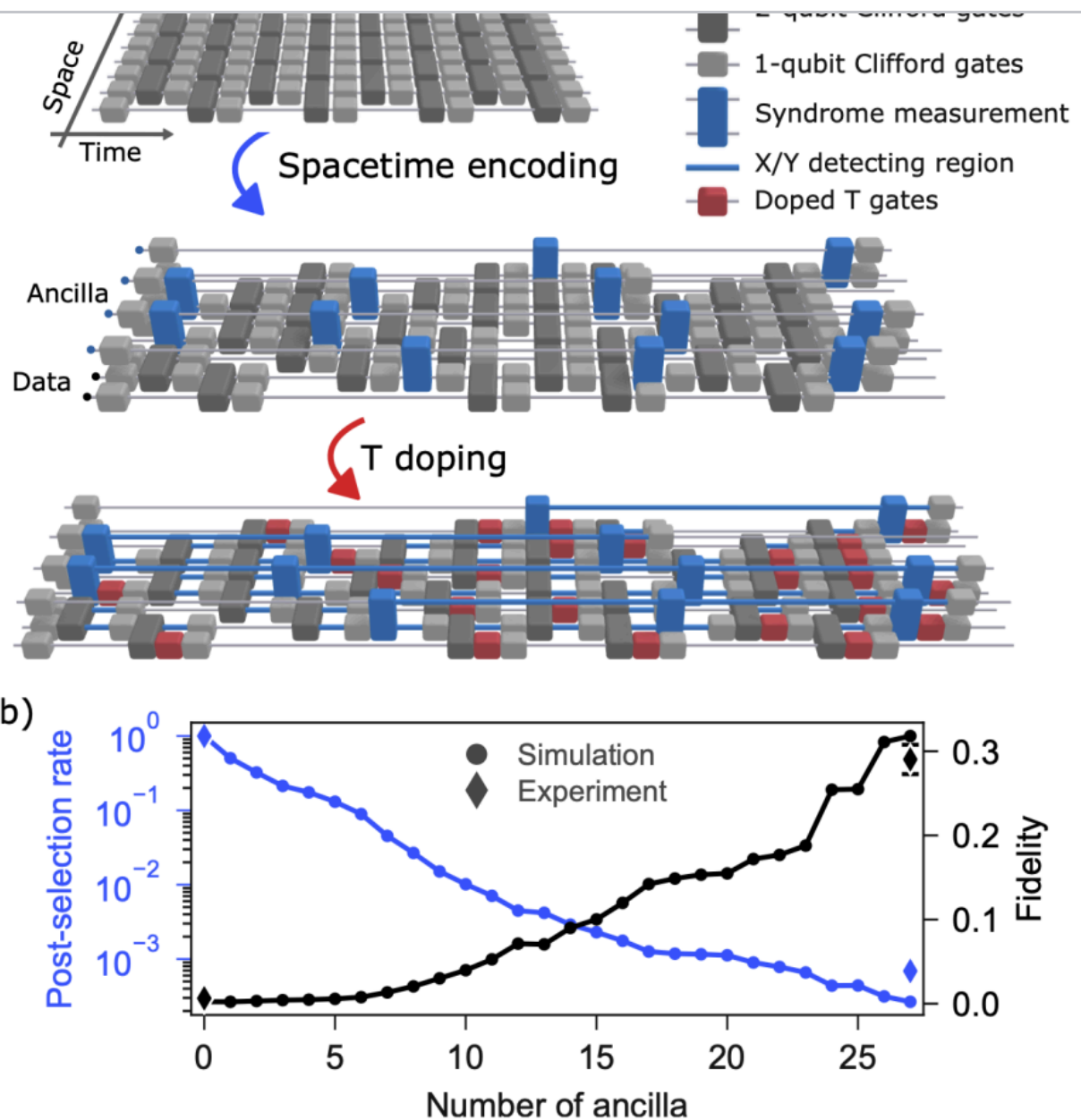
They all use IBM's **Heron R3 quantum processors**, but they have very different focuses. One puts a quantitative lower bound on the fidelity of a difficult quantum circuit. The other two rely on error mitigation, repeat runs, smaller benchmarks, and comparisons with classical simulations that eventually stop agreeing.

If these results hold up, this would mark an important shift in quantum computing that would get us closer to quantum computers being able to do credible work beyond the point where classical machines can check every answer.

Building verification into the circuit

For the first paper, IBM and University of Chicago researchers encoded a 70-qubit circuit across 97 physical qubits. Seventy carried the computation, while 27 ancillas measured checks that could reveal errors.

The calculation began as a **Clifford circuit**, which makes for a good baseline to test quantum hardware because classical computers can simulate it efficiently. The researchers then added 468 T gates, which rotate a qubit's phase by 45 degrees and make the circuit far more difficult to reproduce classically without interfering with its error checks.



Credit: IBM/University of Chicago

When the ancilla measurements produced a nonzero syndrome, the researchers discarded the run, a process known as postselection. The 70-qubit circuit contained 2,415 entangling CZ gates and generated 2,051 postselected samples in 16.1 minutes.

The error checks also gave the researchers enough information to establish a fidelity lower bound of 0.284 with 95% confidence. Fidelity measures how closely the state produced by the machine matches the state it was supposed to produce. The paper describes this as a device-dependent certificate: It uses information from the hardware's checks rather than proving correctness solely from the output samples.

University of Chicago, says in the announcement.

The paper argues that simulating this circuit family is computationally hard under standard complexity assumptions. For the specific experiment, the researchers analyzed leading tensor-network and stabilizer-based approaches and estimated that simulating the complete circuit would be impractical with current versions of those methods. The paper explicitly leaves open the possibility that another classical algorithm could exploit its structure.

It's worth stressing that what we're talking about here isn't a 70-qubit fault-tolerant computer. The circuit detects errors and throws out affected runs. It does not correct errors in real time. This error-detection method also comes at a cost. For the encoded Clifford benchmark, postselection reduced the effective sampling rate by a factor of 860, but the authors note that "this slowdown is tolerable for superconducting systems and still leaves them comparatively faster than trapped-ion and neutral-atom systems."

When classical simulations stop agreeing

For the second paper, Qedma used IBM's processors to simulate a two-dimensional **Floquet Ising model**, which describes how a model magnet behaves when it is hit with a repeating sequence of external pulses.

These simulations covered systems with both 51 and 74 qubits. At earlier points in the calculation, the classical and quantum results agreed. But as the simulation continued, the classical techniques tested by the team either stopped converging or started returning conflicting answers. The error-mitigated quantum calculation, however, continued to show persistent oscillations.

Qedma used an unbiased error-mitigation technique during the earlier cycles, overlapping with the window in which classical verification remained possible. To reach later cycles, it switched to a more scalable technique that doesn't carry the same theoretical guarantee.

The researchers then repeated selected cycles of the 51-qubit calculation on Quantinuum's H2 and Helios trapped-ion processors, which use very different hardware and have different error characteristics from IBM's superconducting processors.

selected cycles on a different kind of quantum computer, combined with the unbiased early-cycle results, provides strong corroboration.

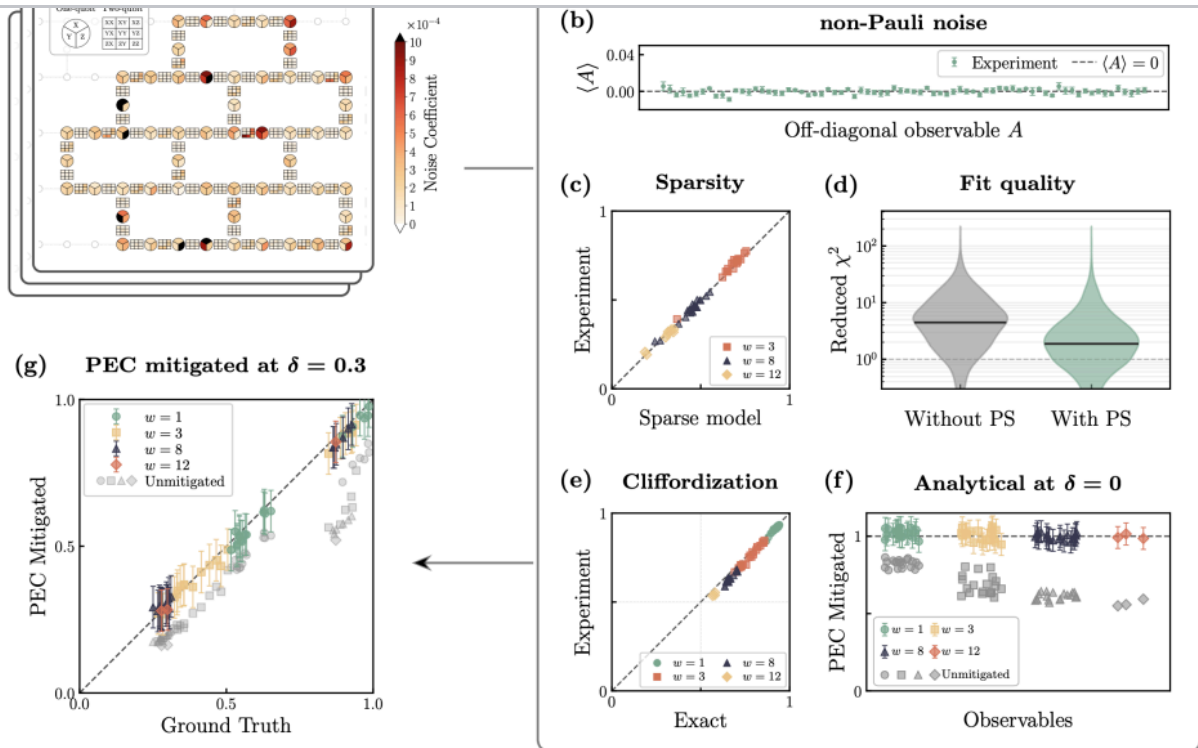
The core idea here is that getting similar results from different quantum architectures makes it less likely that the observed oscillations came from one machine's errors. The paper does hedge a bit, though. It describes the result as a practical classical frontier, not an asymptotic, complexity-theoretic separation, and notes that the limited number of system sizes and their different shapes prevent a controlled extrapolation to an infinitely large system.

Trust through error mitigation

Meanwhile, for the third paper, Algorithmiq and IBM used 56 qubits to measure an operator Loschmidt echo, which is basically a way to track how information moves through a deliberately heterogeneous quantum system.

Algorithmiq sees possible real-world relevance here. Catalysts and battery electrolytes contain interfaces, impurities, and local variations that affect how information, energy, and particles move through them. Sabrina Maniscalco, Algorithmiq's co-founder and CEO, describes these materials as "messy, disordered, and irregular."

But the experiment didn't simulate a particular catalyst or battery. Instead, the team varied the local fields across the qubit lattice, creating fast and slow pathways for information to spread. This gave the researchers a test case for the heterogeneous dynamics that Algorithmiq believes future materials and chemistry simulations will need to handle.



Credit: IBM/Algorithmiq

Different classical approximations agreed with the quantum calculation in different parts of the parameter range, but none returned a reliable answer across the full range. IBM repeated the calculation on different processors, changed gate timings and calibrations, and injected additional noise. After error mitigation, the results remained consistent.

Maniscalco calls this a demonstration of quantum advantage and says related operator Loschmidt echo benchmarks have withstood eight months of challenges on the community-led Quantum Advantage Tracker. In this case, “advantage” doesn’t mean the quantum result was checked against a known answer. What’s worth noting is that the quantum estimate remained stable, while the classical methods the team tested failed to provide a consistent one.

The paper describes the quantum estimate as the “most credible” among the approaches considered.

The paper also distinguishes between two levels of validation. Its formal error-bounded technique is demonstrated on shallower circuits, where classical verification remains possible. The harder 56-qubit calculation relies on heuristic global rescaling, which the researchers tested by deliberately changing the noise affecting the circuit.

It's worth noting, too, that none of the papers excludes every classical algorithm that could be developed for these problems.

"Proving a null result is extremely hard," Gambetta acknowledges.

The community-led **Quantum Advantage Tracker** publishes circuits for all three problem families, along with selected quantum and classical results, and lets researchers challenge them with better classical approaches. It does not currently present the complete headline results from all three papers as active entries.

What's next

Now that IBM has made its case, other researchers get their turn. And when it comes to quantum computing announcements that use the words "quantum advantage," some degree of controversy **typically follows**.

TNS



Before joining The New Stack as its senior editor for AI, Frederic was the enterprise editor at TechCrunch, where he covered everything from the rise of the cloud and the earliest days of Kubernetes to the advent of quantum computing....

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